

MP5 MiniClaw

From Wrong Requirements to Honest Design

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The Brief

What Jordan asked us to build

Jordan Chen Design Brief — MP1

Compete with Hiwonder's BigClaw at RobotExpo 2026

Envelope: $92 \times 46 \times 55$ mm

Grip target: close on a ~ 25 mm object

Thumb-wheel turns: 2–3 from open to closed

Grip force: 5–8 N

Material: FDM PLA (Prusa MK4S)

Part count: < 15 parts

Cost: $< \$3$ /unit at 500-unit scale

Note: brief states no jaw opening — team set ≥ 40 mm as a self-imposed target

"The BigClaw outsells us 3-to-1. Your MiniClaw has to feel just as capable in a package about 20% smaller."

— Jordan Chen, Engineering Manager

Where We Ended Up

Final MiniClaw design vs. targets (MP1 + team)

Specification	Target	Final Design	Status
Jaw opening	≥ 40 mm (team)	40.13 mm	Met
Envelope	$92 \times 46 \times 55$ mm	$92 \times 46 \times 55$ mm	Met
Thumb-wheel turns	2–3	0.28	Renegotiated
Gear stress SF	≥ 2.0	2.14	Met
Part count	< 15	25	Over target
Trans. angle margin	$> 0^\circ$ above 40°	21.3° margin	Met

The Renegotiation Story

GOAL & DIRECTION

Why we accepted 0.28 turns instead of 2–3

Architecture B (compound spur)

$N = 16 \rightarrow 1.24$ thumb-wheel turns

Center distance: 126 mm total

Exceeds 92 mm housing by 34 mm

Part count: 29 (5 gears)

Packaging: DOES NOT FIT

Architecture A (single spur pair)

$N = 1 \rightarrow 0.28$ thumb-wheel turns

Center distance: 20 mm

Fits easily in 92 mm housing

Part count: 25 (2 gears)

Packaging: SOLVED

When Context Was Missing

One document upload corrected the entire design basis

Parameter	AI Without Context	After MP1 Brief Upload
Jaw opening target	40 mm	25 mm
Input range	0°-46°	0°-28°
Link lengths	L1=40, L2=20	L1=12, L2=26
Displacement	20.3 mm/side	12.5 mm/side

Lesson learned:

The AI will confidently build on wrong assumptions if you don't provide the requirements document. Context management isn't optional.

Also missing early: Haben's repo was private (403) until made public. Yoel's MP4 Part A arrived late — his linkage (24 mm jaw, $\mu=57^\circ-119^\circ$) confirmed Haben's design as the right selection.

Tool	Used For
Devin (Cognition AI)	Part A notebooks, Part B templates, plots, stress analysis, 12+ PRs
GitHub Copilot (agent)	Haben's Part A, David's MP3 gear CAD
MiniClaw Gear DFM Skill	Yoel's MCP RAG server + ACME corpus (ACME-ENG-001, MFG-002)
matplotlib / numpy	Comparison plots, linkage simulation, drive-train sketch
GitHub	Version control, code review, team integration

Live demo coming up: animated four-bar simulation + `compute_transmission_angle()` on perturbed geometry

The Pin Clearance Loop

Human × AI = something neither would have produced alone

David asked: "What bore diameter gives a reliable sliding fit after FDM tolerance?"

Bore Ø (mm)	Designed	FDM Offset	Post-FDM	Verdict
3.20	0.20 mm	-0.20 mm	0.00 mm	INTERFERENCE
3.30	0.30 mm	-0.20 mm	0.10 mm	Tight
3.40	0.40 mm	-0.20 mm	0.20 mm	RELIABLE FIT

Human: FDM tolerance data from MP1
AI: systematic clearance stack calculation

Result: bore increased to Ø3.40 mm across all 10 joints. No reaming needed.

The Wrong-AI Moment

EVALUATION & TRUST

What the AI claimed

Designed David's linkage targeting:

- 40 mm jaw opening
- 0° – 46° input range
- $L_1=40$, $L_2=20$ link lengths

Complete design with position equations, transmission angle analysis, motion plots, and hand calculations.

Everything checked out internally.

What was actually true

David's MP1 Part B design used:

- ~25 mm jaw opening (to grip a ~25 mm object)
- 0° – 28° input range
- $L_1=12$, $L_2=26$ link lengths

The AI had never seen David's MP1 work. It guessed the specs and built a plausible-looking but wrong design.

How we caught it

David compared the AI's output against his original design brief.

The numbers didn't match.

One document upload fixed everything.

Mechanism: the human knew the requirements. The AI didn't.

Context management isn't optional — it's the difference between a correct design and a plausible-looking wrong one.

Per-Subsystem Trust Assessment

EVALUATION & TRUST

Subsystem	Flag	Print Tomorrow?	Key Issue
1. Linkage	Needs work	YES	Strongest analytically (3-path verified)
2. Drive Train	Needs work	YES	SF = 2.14, but 0.28 turns untested ergonomically
3. Jaw Arms	Unknown	NO	No CAD exists
4. Housing	Unknown	NO	No CAD exists
5. Input Wheel	Unknown	NO	No CAD exists
6. Pins	Needs work	YES	Ø3.40 bore designed; test one joint first

Bottom line: Not ready to print a complete gripper. Three subsystems are analytically solid; three have no CAD. This is an honest assessment.

What We'd Do Differently

1. Shared team repo from Day 1

Cross-repo integration (cloning Haben's private repo, cloning Yoel's repo, extracting parameters) added friction. A single shared workspace from the start would have eliminated the 403 errors, the manual data extraction, and the integration PRs.

2. Build the MCP tool call earlier

Having `compute_transmission_angle()` live during linkage selection (Loop 1) would have made the comparison interactive instead of static. The tool we demo today should have existed in Week 4.

Live Demo

Yoel Tesfatsion

Part 1: Dynamic Simulation

Haben's four-bar linkage sweeping -50° to $+50^\circ$

Jaw opening from 0 to 40 mm and back

Transmission angle staying in workable band

Real-time data display

Part 2: Live Tool Call

`compute_transmission_angle(L3=32)`

vs. our design ($L3=29$)

Shows angle drops with 3 mm change

Tool catches bad geometry instantly

```
$ Terminal
```

```
$ python3 linkage_simulation.py
```

```
$ python3 transmission_angle_check.py
```

```
--- Haben's design ---
```

```
μ range: 61.3° – 91.3°
```

```
Margin: 21.3° above 40°
```

```
--- Perturbed: L3=32 ---
```

```
μ range: 56.0° – 84.1°
```

```
Margin: 16.0° above 40°
```

Questions?

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Goal & Direction	David Ricciotti
Context Management	Haben Berhe
Tools & Integration	Yoel Tesfatsion
Centaur Engineering	Haben Berhe
Evaluation & Trust	David Ricciotti

Thank you!